

Reward Function and State Space

Octopus CXL Memory Pooling — RL Redesign

Pulak Mehrotra

Notation

m	number of MPDs in the pod
$[m]$	set $\{1, \dots, m\}$
$N(i)$	MPDs reachable from host i
$H(j)$	hosts connected to MPD j
$\deg(h)$	degree of host h , i.e. $ N(h) $
$c_j(t)$	current allocated load on MPD j at time t (GB)
$\ell_h(t)$	current CXL load sourced by host h at time t (GB)
$\mathcal{V}_j(t)$	VMs currently allocated on MPD j at time t
$\text{mem}(v)$	allocated memory of VM v (GB)
$\text{end}(v)$	predicted departure time of VM v ; computed from trace during training, and from a lifetime prediction
x	memory request of the arriving VM (GB)
$\mathbf{w}$	allocation weight vector, $w_j \geq 0$, $\sum_{j \in N(i)} w_j = 1$, $w_j = 0$ if $j \notin N(i)$
W	lookahead window (timesteps; target 150–300)
D_{pod}	total pod DRAM capacity (GB); used for normalisation
λ	Reward-B global-term weight ($\lambda < 1$; start 0.1–0.3)
d_{max}	maximum host degree in the topology
h	current hour of day (UTC)

1 Reward Function

1.1 Departure Relief

Define the time-weighted departure relief on MPD j over the lookahead window $[t, t+W]$:

$$D_j(t, W) = \sum_{\substack{v \in \mathcal{V}_j(t) \\ \text{end}(v) \leq t+W}} \text{mem}(v) \cdot \left(1 - \frac{\text{end}(v) - t}{W}\right) \quad (1)$$

Each VM’s contribution is weighted by how soon it departs: a VM departing at $t+1$ receives weight ≈ 1 (near-certain relief); a VM departing at $t+W$ receives weight ≈ 0 . VMs departing after $t+W$ do not contribute to D_j and instead constitute the sticky load (Section 2).

1.2 Projected Load

The projected load on MPD j before placement is:

$$\hat{c}_j(t) = \frac{c_j(t) - D_j(t, W)}{D_{\text{pod}}} \quad (2)$$

After the agent applies allocation $\mathbf{w}$, the projected load on each reachable MPD $j \in N(i)$ is updated:

$$\hat{c}_j^+(t) = \hat{c}_j(t) + \frac{w_j \cdot x}{D_{\text{pod}}} \quad (3)$$

1.3 Reward

Reward A: local projected-peak baseline.

$$R_A(t) = - \max_{j \in N(i)} \hat{c}_j^+(t) \quad (4)$$

This is the default baseline reward. It penalises the worst projected load among the MPDs the current host can directly influence, keeping the signal local and easy to interpret.

Reward B: local projected peak plus coarse global term.

$$R_B(t) = - \max_{j \in N(i)} \hat{c}_j^+(t) - \lambda \cdot \max_{j \notin N(i)} \hat{c}_j(t) \quad (5)$$

Reward B is an ablation variant that adds a coarse global-stress term. The first term is identical to Reward A. The second term, down-weighted by λ , penalises the projected peak among unreachable MPDs. Its purpose is to test whether coarse pod-level stress helps value estimation beyond the local baseline.

All values are normalised by D_{pod} . Reward A lies in $(-1, 0]$. Reward B lies in $-(1 + \lambda), 0]$. This keeps both reward variants on a stable scale across traces and pod sizes.

2 State Space

2.1 Design Principle

The state is shared across Reward A and Reward B. Its main job is to describe, for each reachable MPD, both *(i)* how much load is currently there and how much of that load is likely to drain soon, and *(ii)* how contested that MPD is by nearby hosts. This keeps the baseline design local and interpretable while exposing the neighborhood structure that greedy does not model explicitly.

2.2 Per-MPD Features

The following features are computed for each MPD slot $k = 1, \dots, d_{\text{max}}$, zero-padded for slots $k > |N(i)|$:

$$\phi_k^{(1)} = \frac{c_{j_k}(t)}{D_{\text{pod}}} \quad \text{current allocated load on MPD } j_k \quad (6)$$

$$\phi_k^{(2)} = \frac{D_{j_k}(t, W)}{D_{\text{pod}}} \quad \text{time-weighted departures within } W \text{ steps} \quad (7)$$

$$\phi_k^{(3)} = \frac{S_{j_k}(t)}{D_{\text{pod}}} \quad \text{sticky load (departs after } W \text{ steps)} \quad (8)$$

$$\phi_k^{(4)} = \mathbf{1}[j_k \in N(i)] \quad \text{connectivity mask / action validity} \quad (9)$$

$$\phi_k^{(5)} = \frac{P_{j_k}(t)}{D_{\text{pod}}} \quad \text{neighborhood pressure on MPD } j_k \quad (10)$$

$$\phi_k^{(6)} = Q_{j_k} \quad \text{neighbor scarcity around MPD } j_k \quad (11)$$

where the sticky load is defined as:

$$S_j(t) = \sum_{\substack{v \in \mathcal{V}_j(t) \\ \text{end}(v) > t+W}} \text{mem}(v) \quad (12)$$

and the two neighborhood-contention features are defined as:

$$P_j(t) = \sum_{h \in H(j)} \ell_h(t) \quad \text{total current host pressure from hosts attached to } j \quad (13)$$

$$Q_j = \frac{1}{|H(j)|} \sum_{h \in H(j)} \frac{1}{\deg(h)} \quad \text{average scarcity of alternatives among hosts attached to } j \quad (14)$$

Note that $\phi_k^{(1)} = \phi_k^{(2)} + \phi_k^{(3)}$ up to normalisation: current load decomposes exactly into draining and sticky components. The agent is therefore given the full decomposition of $\hat{c}_j$ directly rather than having to infer it from a single load snapshot. The added $\phi^{(5)}$ and $\phi^{(6)}$ terms provide a simple 2-hop summary of which reachable MPDs are likely to be needed by neighboring hosts.

2.3 Global Context Features

$$\psi^{(1)} = \frac{\max_{j \in [m]} c_j(t)}{D_{\text{pod}}} \quad \text{global peak load (used only by Reward B)} \quad (15)$$

$$\psi^{(2)} = \frac{x}{D_{\text{pod}}} \quad \text{arriving VM size} \quad (16)$$

$$\psi^{(3)} = \sin\left(\frac{2\pi h}{24}\right) \quad \text{hour-of-day encoding (sin)} \quad (17)$$

$$\psi^{(4)} = \cos\left(\frac{2\pi h}{24}\right) \quad \text{hour-of-day encoding (cos)} \quad (18)$$

2.4 Full Observation Vector

$$s_t = [\phi_1^{(1:6)}, \phi_2^{(1:6)}, \dots, \phi_{d_{\max}}^{(1:6)}, \psi^{(1)}, \psi^{(2)}, \psi^{(3)}, \psi^{(4)}] \in \mathbb{R}^{6d_{\max} + 4} \quad (19)$$

For the AG16x6 topology ($d_{\max} = 8$) this yields a **52**-dimensional observation vector.

2.5 State–Reward Alignment

Table 1: Shared state features used by Reward A and Reward B.

Feature	Reward quantity	Role
$\phi^{(1)}$	c_j/D_{pod}	Starting point for $\hat{c}_j$
$\phi^{(2)}$	$D_j(t, W)/D_{\text{pod}}$	Draining component of $\hat{c}_j$
$\phi^{(3)}$	$S_j(t)/D_{\text{pod}}$	Permanent component of $\hat{c}_j$
$\phi^{(4)}$	Action mask	Enforces topology constraint
$\phi^{(5)}$	$P_j(t)/D_{\text{pod}}$	Neighborhood contention around MPD j
$\phi^{(6)}$	Q_j	Scarcity of alternatives for nearby hosts
$\psi^{(1)}$	$\max_j c_j/D_{\text{pod}}$	Coarse global context for Reward B only
$\psi^{(2)}$	x/D_{pod}	Post-placement update magnitude
$\psi^{(3,4)}$	Hour-of-day	Diurnal demand pattern

2.6 Comparison with v0

Table 2: State space comparison.

Feature	v0	This work
Current MPD load	✓	✓
Connectivity mask	✓	✓
Arriving VM size	✓	✓
Global peak load	✓	✓
Hour-of-day	✓	✓
Departure relief $D_j(W)$	×	✓
Sticky load S_j	×	✓
Neighborhood pressure P_j	×	✓
Neighbor scarcity Q_j	×	✓
Dimensions (AG16x6)	20	52

Relative to v0, the added dimensions come from two sources: future-pressure decomposition (D_j, S_j) and neighborhood-contention signals (P_j, Q_j). Together they tell the policy not only what load may drain soon, but also which reachable MPDs are strategically valuable because they serve a more constrained neighborhood.

3 Hyperparameters

W should be set to at least the median VM lifetime in the trace. The future-knowledge ablation [2] shows that the marginal value of lookahead is low below ≈ 150 steps and rises steeply above that

Table 3: Tunable hyperparameters.

Parameter	Role	Starting value
W	Lookahead window	150–300 steps (12.5–25 hr at 5-min resolution)
λ	Reward-B global-term weight	0.1–0.3

threshold, justifying the lower bound. λ is only used in Reward B. It should be kept small enough that the local term still dominates the reward gradient across a training batch; verify empirically before scaling to 96-host pods.

4 Potential Issues to Validate

This section records the main formulation risks to check during implementation and ablation. These are not claims that the design cannot work; they are the most likely places where the reward/state proposal could fail to beat greedy or could produce misleading training signal.

4.1 Highest-Priority Risks

1. **Reward B’s λ term is only a coarse ablation.** The Reward B term $\max_{j \notin N(i)} \hat{c}_j(t)$ is not reconstructed exactly by the shared state. This is acceptable for an ablation, but Reward B should not be described as a perfectly aligned state–reward design.
2. **Reward B’s λ term may be action-independent at each step.** For a given VM arrival, the agent cannot change the loads of unreachable MPDs. As written, the second reward term may therefore be constant with respect to the current action, which can add critic noise without helping the policy distinguish better placements. This should be checked empirically before relying on it as useful shaping.
3. **Missing lifetime signal for the arriving VM.** The redesign distinguishes draining load from sticky load for already-placed VMs, but the state includes only the arriving VM size x , not the arriving VM’s predicted lifetime. If the new VM itself can be short-lived or long-lived, then the policy is missing a key variable needed to reason about whether the new placement will create future pressure.
4. **No explicit feasibility / overflow term is documented.** The reward penalises projected load but does not explicitly encode a hard penalty for exceeding MPD capacity. If infeasible allocations are blocked by an action mask or safety layer, that should be stated clearly. If not, the policy may optimise the smooth peak surrogate while still learning unsafe behavior near capacity.

4.2 Secondary Risks

1. **Reward ranges should be reported per variant.** Reward A and Reward B have different lower bounds. The text should keep those bounds separate rather than describing a single shared range for all reward variants.
2. **Normalising by D_{pod} may weaken local signal.** The systems objective is driven by per-MPD peaks, but D_{pod} grows with pod size. The same bad local placement can therefore

produce a smaller reward difference in larger pods. A per-MPD capacity scale, or a fair-share normalization, may align better with the actual control objective.

3. **The reward may remain close to a future-aware greedy rule.** The first term still behaves like “minimise the post-placement projected local maximum over reachable MPDs.” That is a reasonable heuristic, but it is not obviously the same as minimising the final trace-wide peak. The formulation should be validated by checking whether greedy, random, and stronger baselines rank correctly under this reward on held-out traces.
4. **Neighborhood contention features need empirical validation.** The new P_j and Q_j signals are intentionally simple summaries of local competition. They may help beat greedy, but they should still be ablated to confirm that they improve decisions rather than just adding noise.
5. **Slot ordering may create a permutation problem.** The observation is written as slot-indexed features $\phi_k^{(1:4)}$. If MPDs are not placed into those slots using a canonical order, then equivalent neighborhoods may appear as different input vectors across episodes, making the MLP learn a harder permutation-sensitive task than intended.

4.3 Clarifying Questions to Resolve

- Is D_{pod} total pod DRAM, total MPD capacity, or a fair-share normalization constant?
- Is the arriving VM’s lifetime known at decision time, or does deployment require a predictor for that quantity as well?
- Are capacity violations prevented by a hard mechanism outside the reward, or should the reward itself teach the policy that infeasible actions are unacceptable?

References

- [1] Huaicheng Li, Daniel S. Berger, Lisa Hsu, et al. Pond: CXL-Based Memory Pooling Systems for Cloud Platforms. In *ASPLOS*, 2023.
- [2] Richard Uzeel. Perfect Future Knowledge. Internal project report, 2026.